📊 Phase 2: RFM Customer Segmentation (Summary)

In this phase, I translated clean data into actionable business intelligence by implementing the **RFM (Recency, Frequency, Monetary)** model. The primary goal was to analyze purchasing behaviors and identify the most valuable customer segments within the Olist ecosystem.

1. Methodology & Research Process

- **Technical Challenges:** Initial analysis suggested nearly zero customer retention. I discovered that this was due to the customer_id being unique to each session. To accurately identify returning customers, I mapped the data to a persistent key: customer_unique_id.
- **Error Elimination:** I identified a "row inflation" issue in the payments table, where installment payments (up to 29 rows per order) were incorrectly inflating transaction counts. I applied the COUNT(DISTINCT order_id) function to obtain a true reflection of purchase frequency.
- **SQL Implementation:** The model was built using advanced CASE WHEN statements to assign customers to segments based on the time since their last purchase (Recency) and their total lifetime spend (Monetary).

2. Key Business Insights

- **The "Leaky Bucket" Challenge:** 97% of the database consists of one-time buyers. This indicates that the company is highly dependent on expensive, continuous customer acquisition through advertising.
- **Hidden Loyalty:** Despite low overall retention, I identified a group of over 2,400 repeat customers, primarily within the "New/Active" and "Occasional" segments.
- **Critical Segments:**
    - **Recent Champions:** A small but elite group of 225 individuals with a 10% retention rate—significantly higher than the company average.
    - **At Risk High-Value:** A crucial group of 473 customers who previously generated high revenue (over 1,000 units) but are currently at risk of total churn.

3. Strategic Recommendations

- **Win-back Campaign:** Launch a personalized "We Miss You" campaign targeting the 473 high-risk customers who have already proven their willingness to spend significant amounts.
- **CRM Automation:** Implement automated post-purchase communication. Data shows that if a customer does not return within 90 days (Recency score 5), the probability of a subsequent purchase drops drastically.
- **Success Analysis:** Conduct a deep dive into the behaviors of the 24 repeat buyers within the "Champions" segment to understand which product categories or delivery standards drive long-term loyalty.

4. Future Optimization

While the RFM analysis was fully implemented in SQL to ensure high performance on the database side, I see significant potential in using tools like **Power BI** to dynamically visualize customer transitions between segments in real-time.